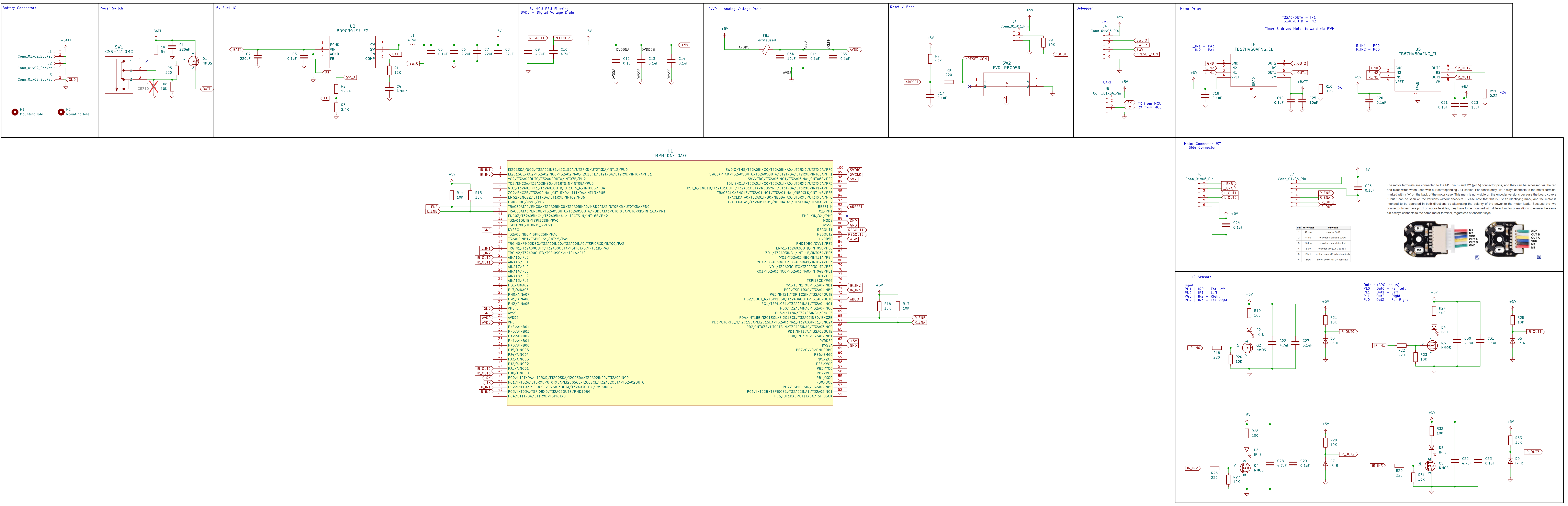
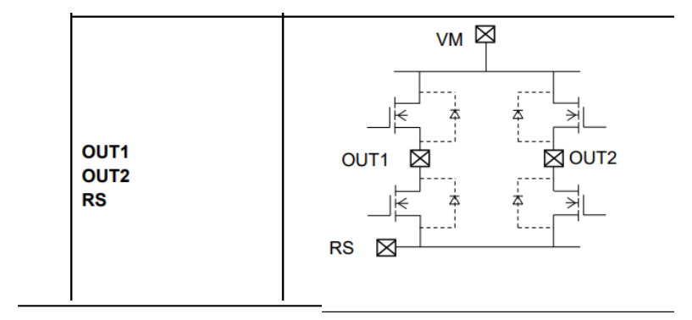


The motor terminals are connected to the M1 (pin 6) and M2 (pin 5) connector pins, and they can be accessed via the red and black wires when used with our corresponding JST cables. For consistency, M1 always connects to the motor terminal marked with a "+" on the back of the motor case. This mark is not visible on the encoder versions because the board covers it, but it can be seen on the versions without encoders. Please note that this is just an identifying mark, and the motor is intended to be operated in both directions by alternating the polarity of the power to the motor leads. Because the two connector types have pin 1 on opposite sides, they have to be mounted with different motor orientations to ensure the same pin always connects to the same motor terminal, regardless of encoder style.

Pin	Wire color	Function
1	Green	encoder GND
2	White	encoder channel B output
3	Yellow	encoder channel A output
4	Blue	encoder Vcc (2.7 V to 18 V)
5	Black	motor power M2 (other terminal)
6	Red	motor power M1 ("+" terminal)

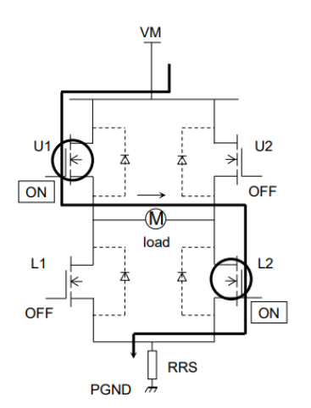
Input:
PU1 | IR0 - Far Left
PU0 | IR1 - Left
PG5 | IR2 - Right
PG4 | IR3 - Far Right

Output (ADC Inputs):
PL0 | Out0 - Far Left
PL1 | Out1 - Left
PJ1 | Out2 - Right
PJ0 | Out3 - Far Right



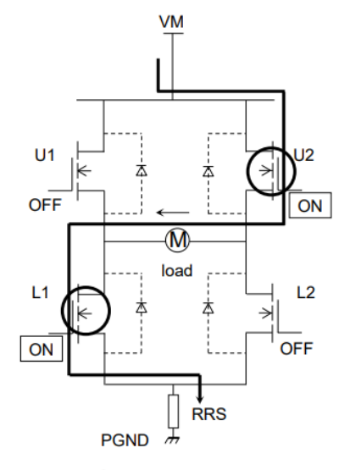
9. Operation Mode of Out

9.1. Forward Rotation



Charge operation
A current flows into the motor coil.

9.2. Reverse Rotation



Charge operation
A current flows into the motor coil.

8. Functional Description

8.1. Input and Output Function

Table 8.1 Input and Output Function

IN1	IN2	OUT1	OUT2	Mode
L	L	OFF (Hi-Z)	OFF (Hi-Z)	Stop Standby mode after t_{stby}
H	L	H	L	Forward
L	H	L	H	Reverse
H	H	L	L	Brake

Note: Current path: Forward rotation (OUT1 to OUT2), Reverse rotation (OUT2 to OUT1)

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